

Quiz

May. 18, 2026

填空题 2分

Given four processes P1 through P4 and three resource types in the operating system: A (5 instances), B (5 instances), C (4 instances). At time T, the Max matrix, Allocation matrix, and Request matrix are shown below.

Is the system in the deadlock state? [填空1] (yes/no)

If the system in the safe state? [填空2] (yes/no)

	Max			Allocation			Request		
	A	B	C	A	B	C	A	B	C
P1	4	1	4	2	1	2	0	0	2
P2	2	4	3	1	2	1	1	1	1
P3	2	1	0	2	0	0	0	0	0
P4	2	2	1	0	1	1	1	1	0

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填空题 3分

A system has 3 resource types:

A with 9 instances

B with 6 instances

C with 8 instances

At time T_0 , the system state is

Process	Allocation (A B C)	Max (A B C)	Available (A B C)
P0	1 0 1	5 4 3	2 2 2
P1	2 1 0	4 3 2	
P2	1 1 1	3 2 2	
P3	0 0 2	2 1 4	

(a) Calculate the Need matrix.

Need[P2] = ([Blank 1] , [Blank 2] , 1)

(b) The system is in a safe state. One safe sequence is:

P0 → [Blank 3] → P3 → [Blank 4]

(c) Then, P1 requests (1, 0, 1). Using the Banker's Algorithm, should this request be granted immediately?

Answer: [Blank 5] (Yes / No)

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填空题 2分

[填空1] I/O and [填空2] I/O are two approaches to performing programmed I/O; the former requires privileged instructions and the latter maps device registers in the physical address space.

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Which of the following statements about polling vs. interrupts are correct?

A

Polling is generally more efficient than interrupts for very slow devices because it avoids context switch overhead.

B

Interrupts allow the CPU to switch to another task while waiting for I/O completion.

C

A hybrid approach can poll for a short time, then fall back to using an interrupt.

D

Polling wastes CPU time when the device is slow, but works well for fast devices.

E

Interrupts are always better than polling for all types of I/O devices.

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Which of the following are true about DMA (Direct Memory Access)?

- A** DMA transfers data directly between I/O device and memory without CPU involvement for each byte.
- B** DMA completely eliminates the need for interrupts.
- C** The CPU must copy each byte of data from the device to memory when DMA is used.
- D** After DMA completes, it raises an interrupt to signal the CPU.
- E** DMA can perform cycle stealing from the CPU but is still more efficient than programmed I/O.

提交

Which of the following statements about modern SSDs (Solid State Disks) are correct?

- ☐ A SSDs have no moving parts, eliminating seek time and rotational latency.
- ☐ B NAND flash can be overwritten directly at the page level without erasing.
- ☐ C 3D NAND stacks memory cells vertically to increase density.
- ☐ D QLC (4-bit per cell) offers higher endurance than SLC.
- ☐ E NVMe SSDs connect via PCIe and support multiple deep command queues (up to 64K).

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